High performance Analog and Mixed signal VLSI System design

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MEDIATEK

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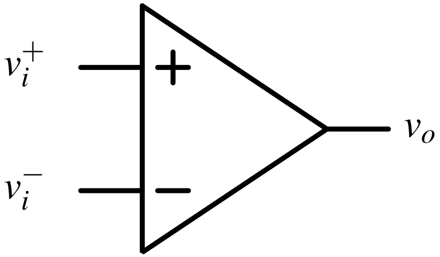
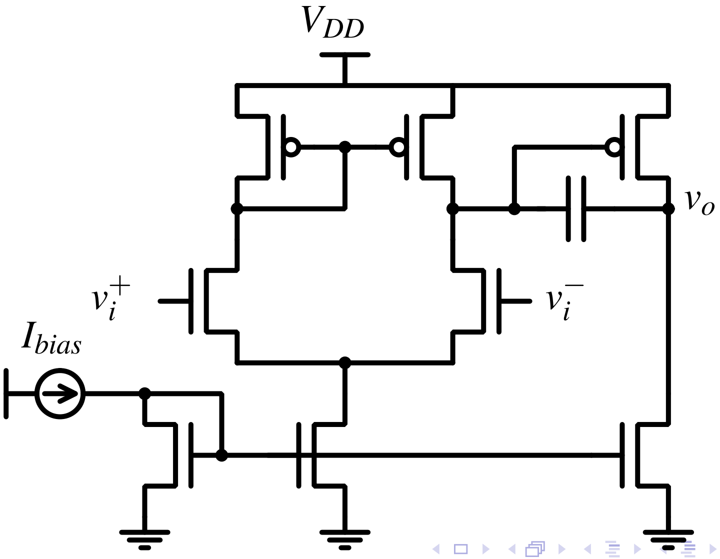
Fully differential amplifiers

Differential amplifiers with single ended output

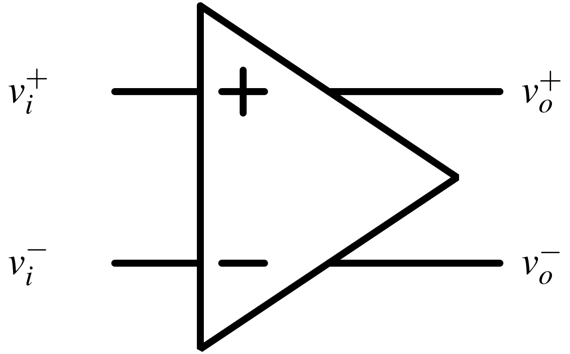
- We have seen the design of high gain multi-stage amplifiers and their compensation for stability.
- All the configurations we have studied had a differential input and a single ended output.





Fully differential amplifiers

- A fully differential amplifier is one in which the input and output are both differential.



- Fully differential configurations are useful in high speed circuits like filters, transcievers etc. where immunity against power supply and ground noise is desired.

Fully differential amplifiers: with resistive loads

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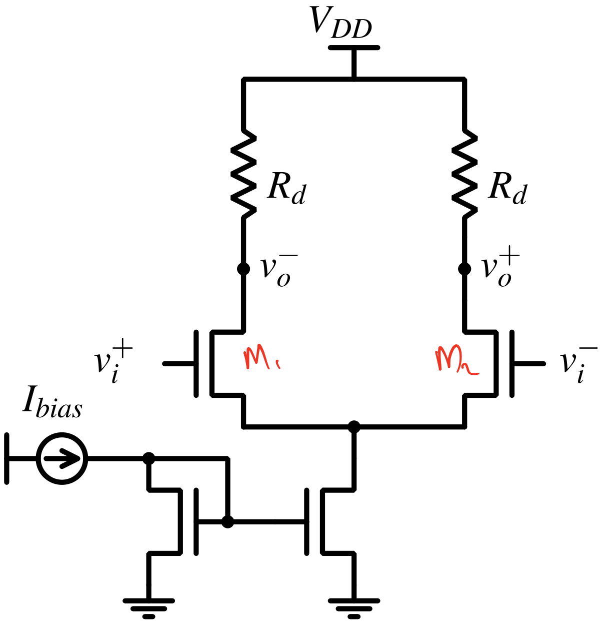

- Designining this circuit for very high gains without increasing the supply voltage is difficult.

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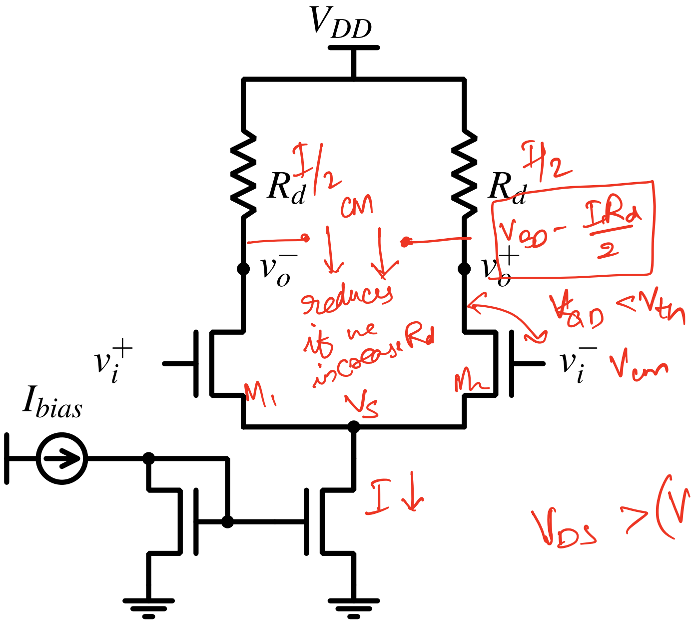

- Desigining this circuit for very high gains without increasing the supply voltage is difficult.
- This architecture is however often used when we need fully differential amplifiers with low gain and very wide bandwidth.

$V_{DS} > (V_{GS} - V_t)$

Fully differential amplifiers: with resistive loads

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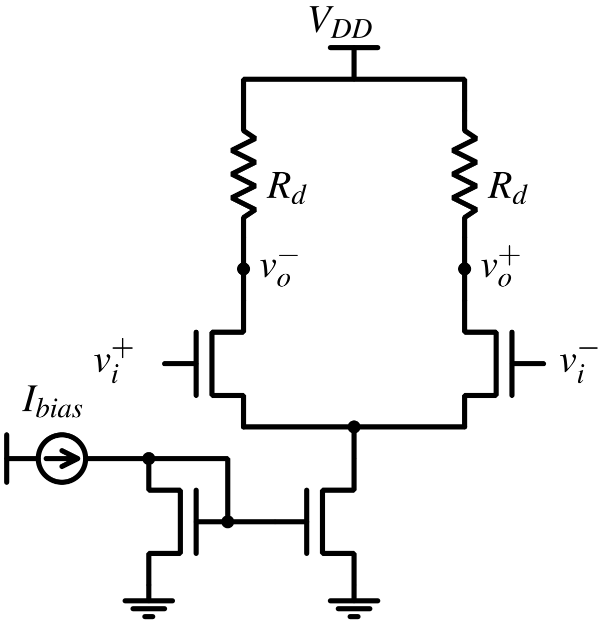

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- This architecture is however often used when we need fully differential amplifiers with low gain and very wide bandwidth.
- *More detailed discussions on this circuit will follow in our later sessions.*

Fully differential amplifier with high gain

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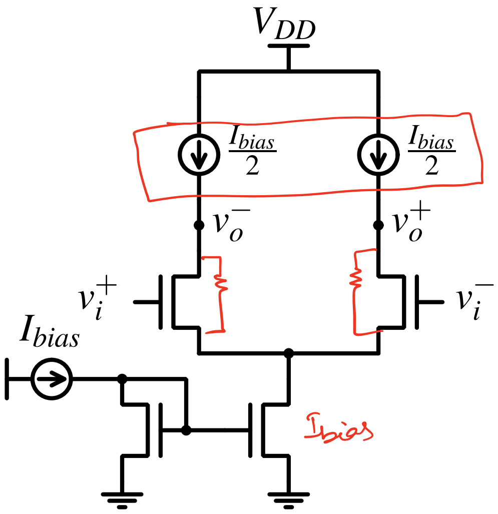

- This circuit will have a gain of $g_m r_o$ where r_o is the output impedance of the NMOS input transistor in saturation.

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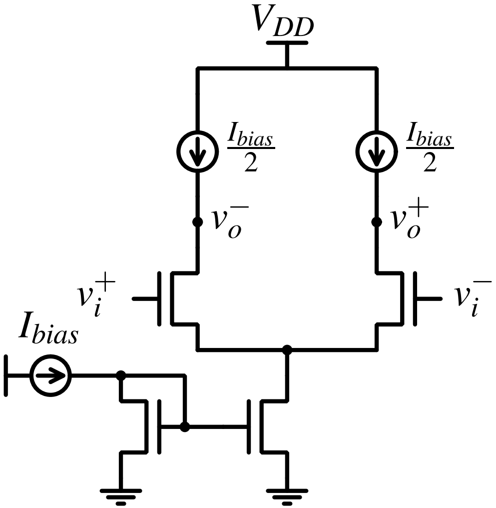

- This circuit will have a gain of $g_m r_o$ where r_o is the output impedance of the NMOS input transistor in saturation.
- How can we realize such a circuit using transistors?

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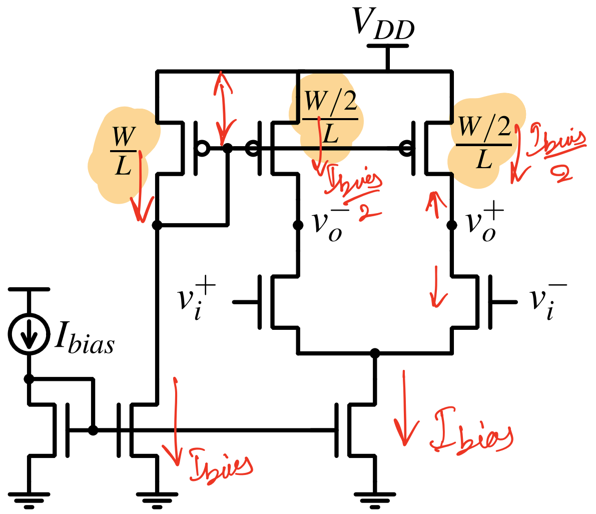

- This circuit will have a gain of $g_m r_o$ where r_o is the output impedance of the NMOS input transistor in saturation.
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- The gain = $g_m(r_{on} \parallel r_{op})$

Fully differential amplifier with high gain

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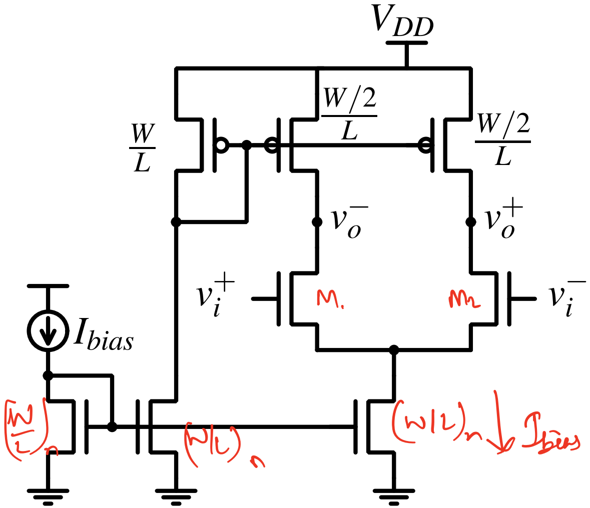
V_{os} depends on $\lambda \rightarrow \lambda$ variation?
mismatch



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- How can we realize such a circuit using transistors?
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- Will this work?

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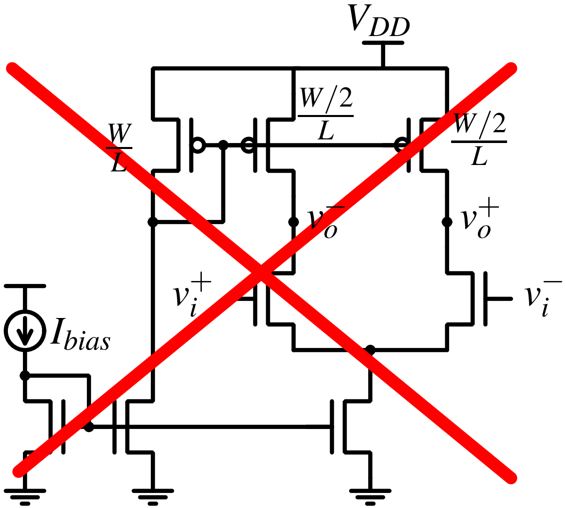
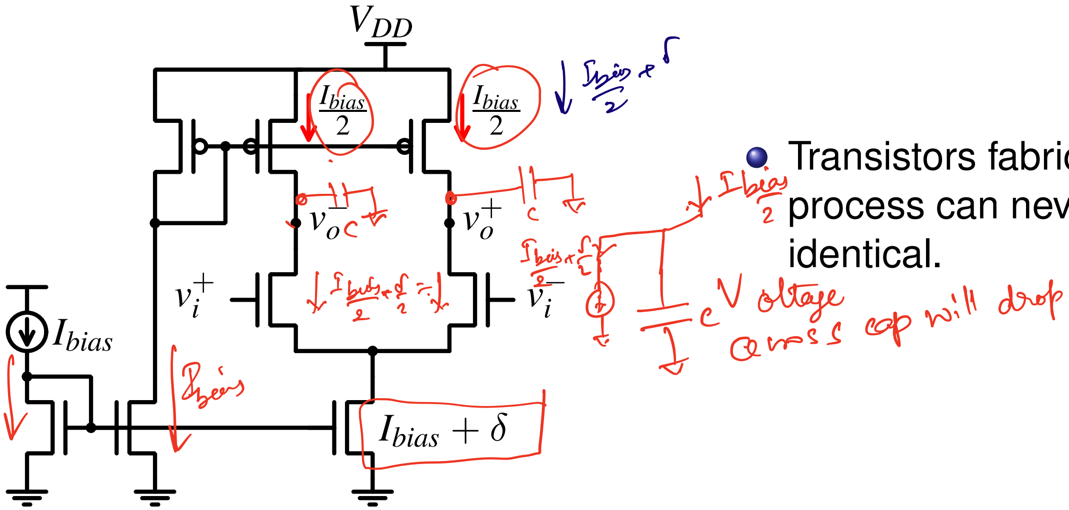


- This circuit will have a gain of $g_m r_o$ where r_o is the output impedance of the NMOS input transistor in saturation.
- How can we realize such a circuit using transistors?
- The gain = $g_m(r_{on} \parallel r_{op})$
- Will this work?
- What is the problem with this circuit?

Fully differential amplifier with current source loads

- To understand why this circuit will not work practically, we need to see what happens in the presence of mismatch.



- Transistors fabricated in any process can never be exactly identical.

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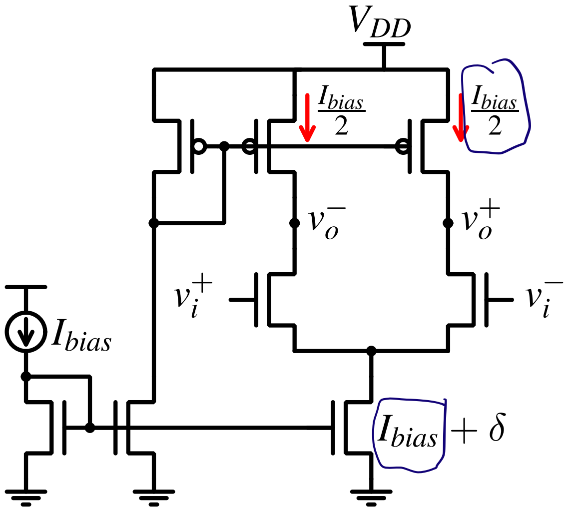
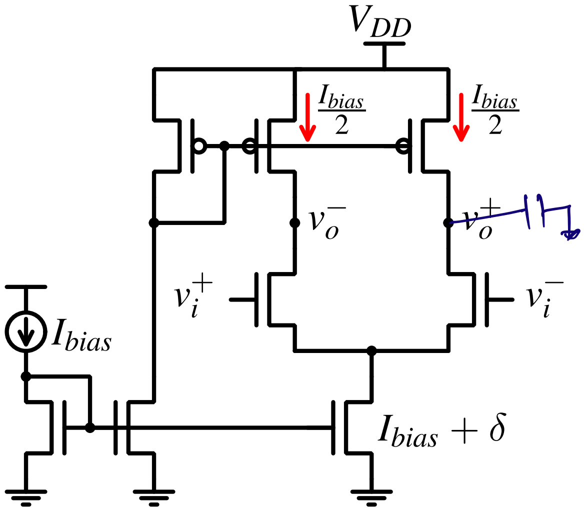
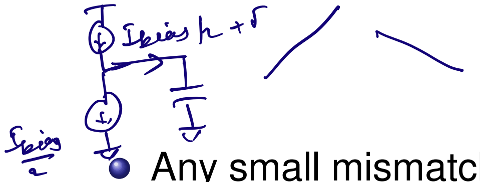


- Transistors fabricated in any process can never be exactly identical.
- The circuit shown will work correctly only if the tail current is **exactly** two times the current source load value.

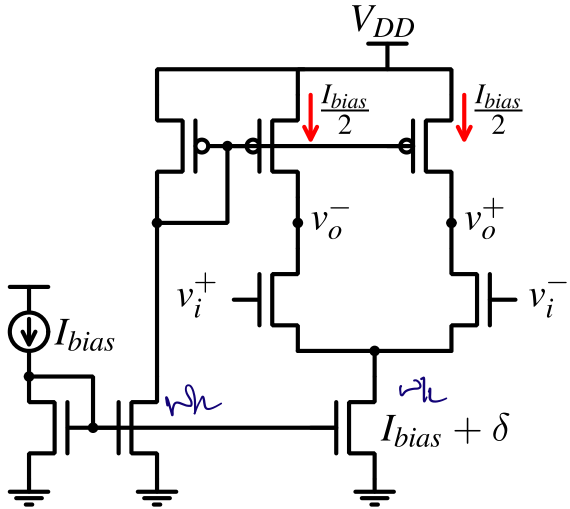
Fully differential amplifier with current source loads





- Any small mismatch (however arbtirarily small it may be) will result in the output either charging up (or down) till the current source load (tail ~~current~~ i/p pair source) transistor goes out of saturation.
- Once that happens the gain will decrease and the differential amplifier will not work as expected.

Fully differential amplifier with current source loads



Fully differential amplifier with current source loads

- How can we solve this problem and design a fully differential amplifier with high gain?

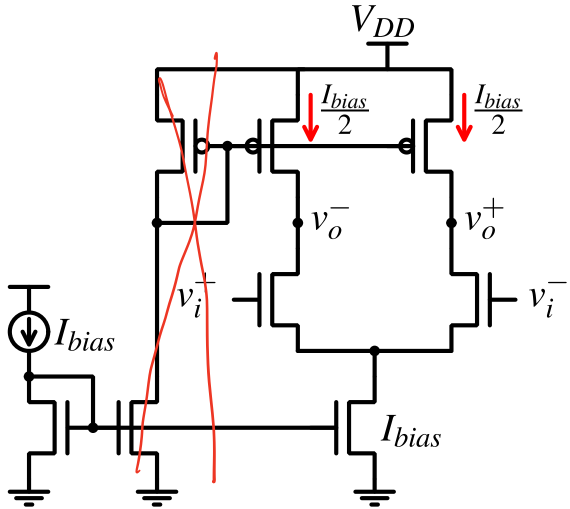
Fully differential amplifier with current source loads

- How can we solve this problem and design a fully differential amplifier with high gain?
- The way to do this would be to somehow actively track the operating point of the transistors and bias the current sources accordingly to keep the transistors in saturation region.

Fully differential amplifier with current source loads

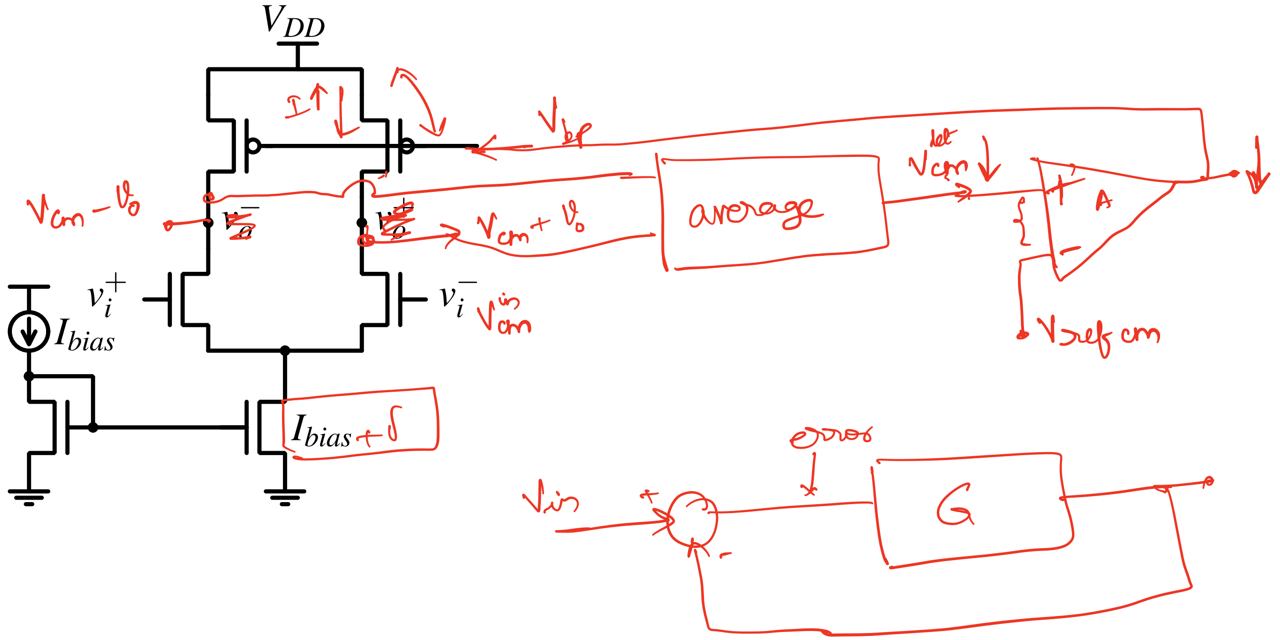
- How can we solve this problem and design a fully differential amplifier with high gain?
- The way to do this would be to somehow actively track the operating point of the transistors and bias the current sources accordingly to keep the transistors in saturation region.
- Note that when there is a mismatch, the output common mode voltage drifts up or down (depending on the direction of the mismatch) till one of the transistors goes out of saturation.
- If we sense the output common mode somehow and adjust the current source bias to keep the common mode at the desired value, we would have achieved our objective.

How output common mode changes with current mismatch

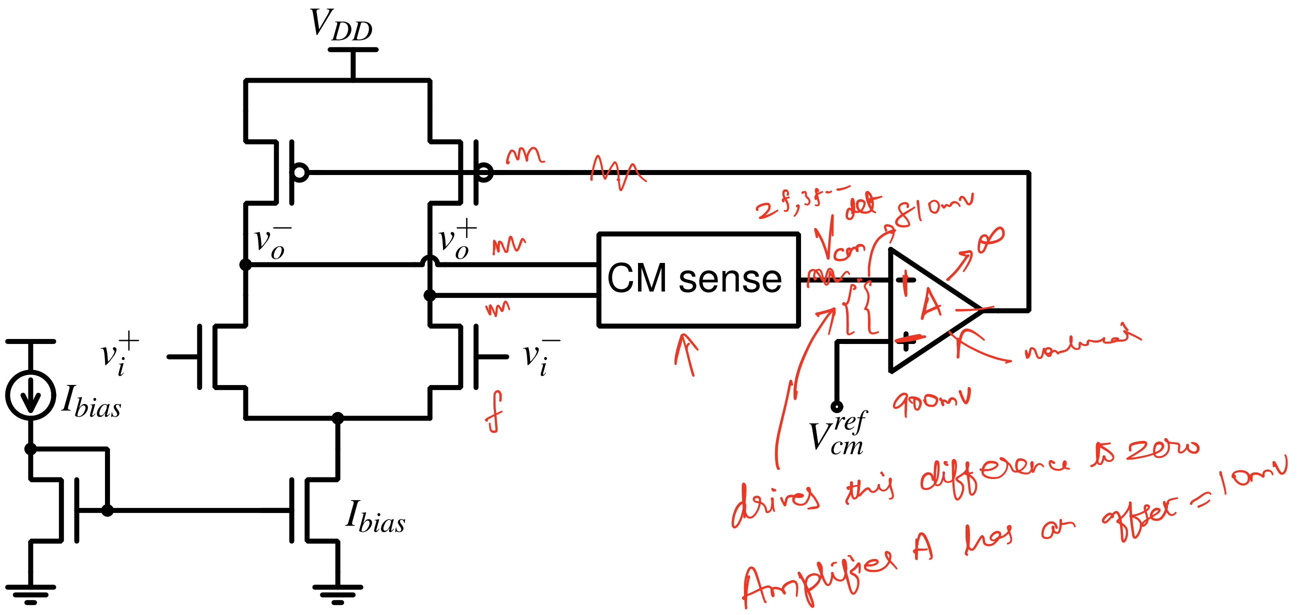


If we have mismatch → o/p common mode drifts (UP/DN)

Common mode feedback

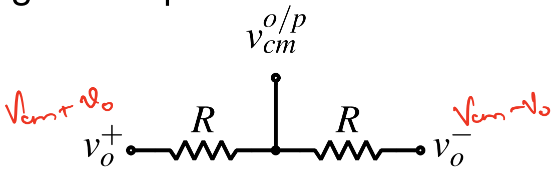


Fully differential amplifier with common mode feedback



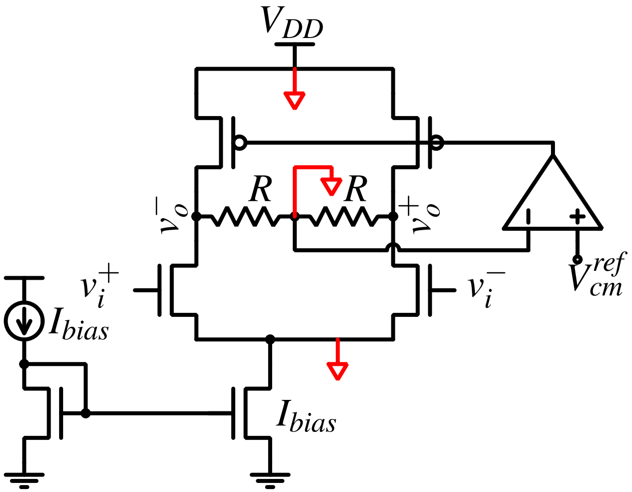
How do we detect the common mode of the output?

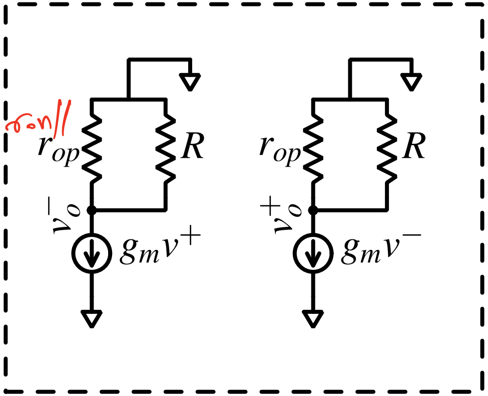
- One simple way of detecting the output common mode is by using a resistor network.



- This simple network of 2 resistors works very well to detect common mode. However, it affects the gain of the amplifier for differential input signals.

Gain with resistive common mode detector



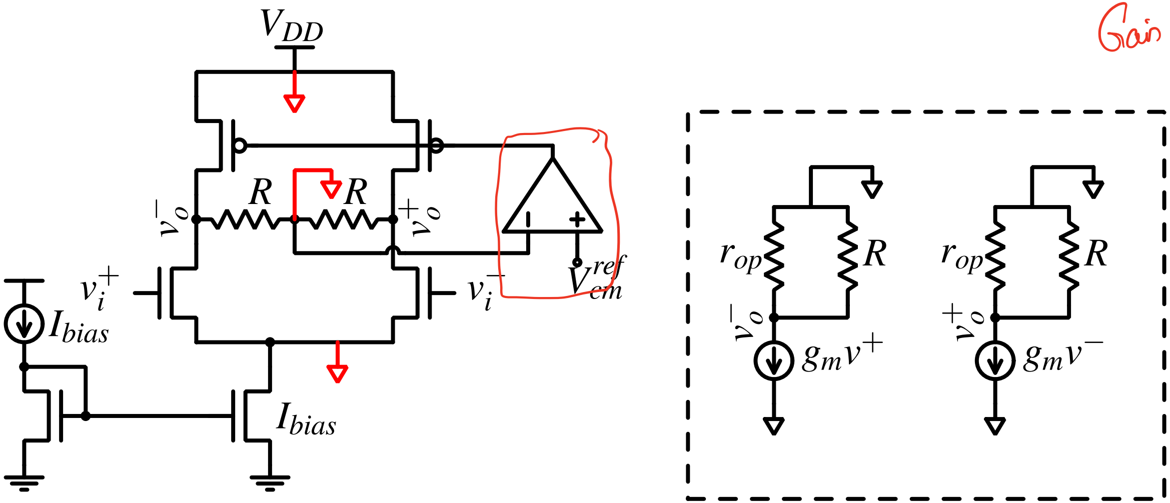


- AC ground (small signal ground) is shown in red.
- As can be seen from the small signal equivalent circuit, the output impedance is reduced to $r_{op} \parallel R$.

Gain with resistive common mode detector

Gain = $g_m R$

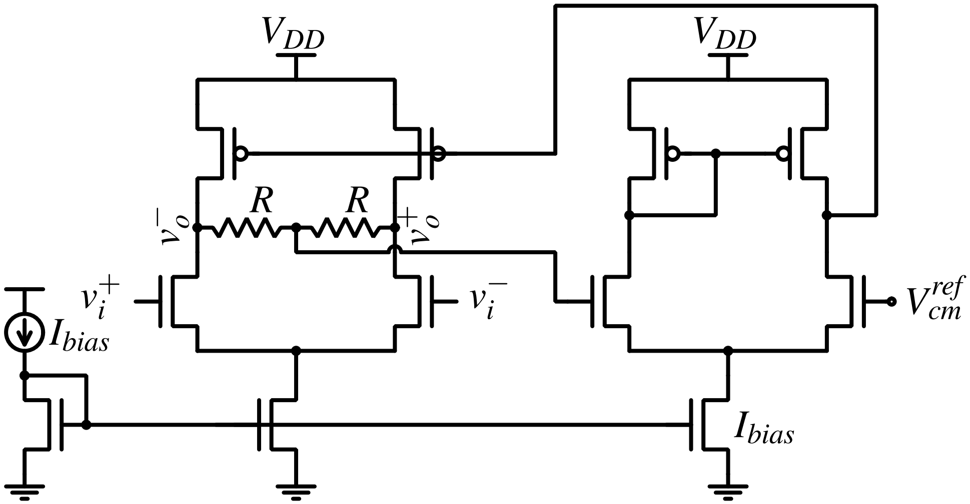
$r_{on}, r_{op} \gg R$



- This reduces the gain of the amplifier, unless very large values of resistors are used. Large values of resistors ($>> r_o$ of transistors is saturation region) is often not practical.

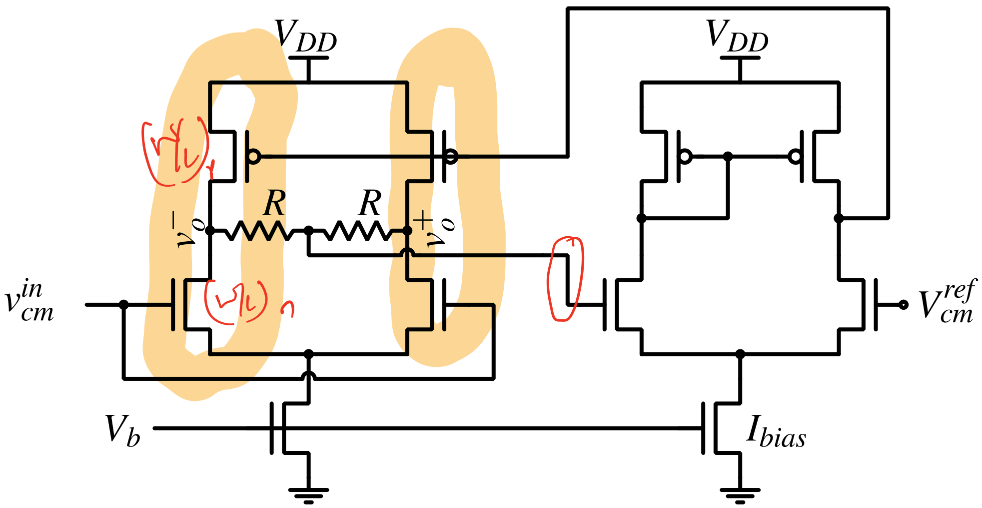
Fully differential amplifier with resistive CM detector

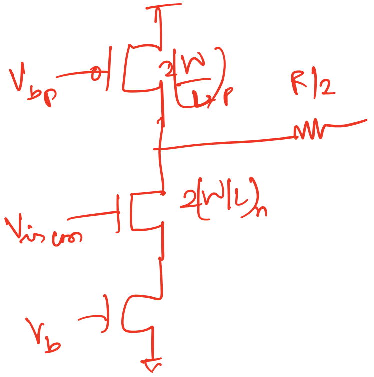
- Apart from gain reduction, there is another serious issue - Stability of the common mode feedback loop.
- Let us look at the transistor level circuit of the common mode feedback loop.



Stability of common mode feedback loop

- The common mode equivalent of this circuit is same as the two stage opamp with differential input and single ended output.





equivalent ~~common~~ common mode circuit.

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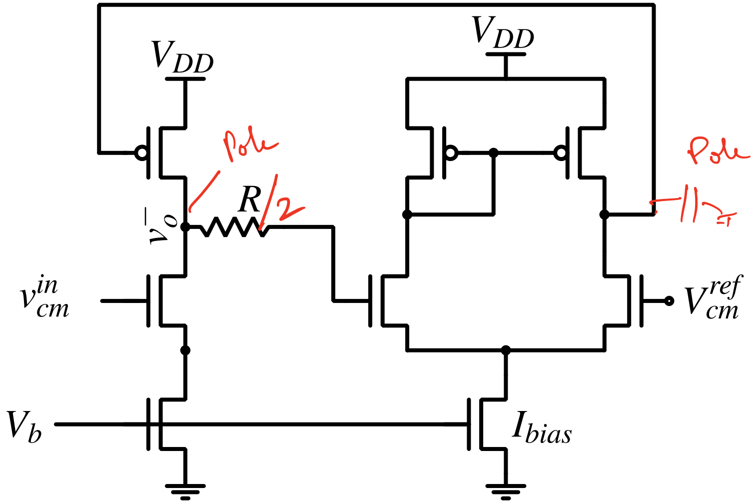
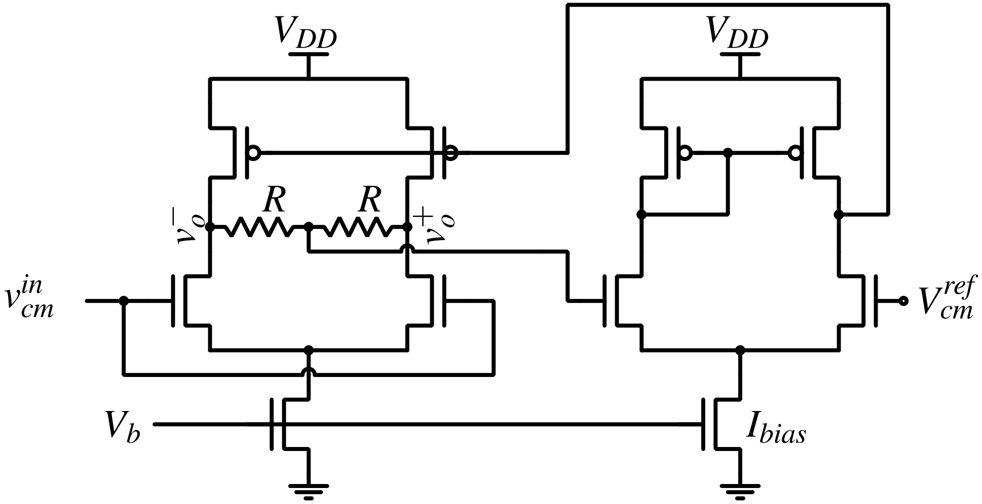


Stabilizing the common mode feedback loop

- Can we stabilize the Common mode feedback loop using a miller capacitor?



- Using a miller capacitor will stabilize the loop, but this capacitor will reduce the bandwidth of the amplifier for differential signals which is not desirable.

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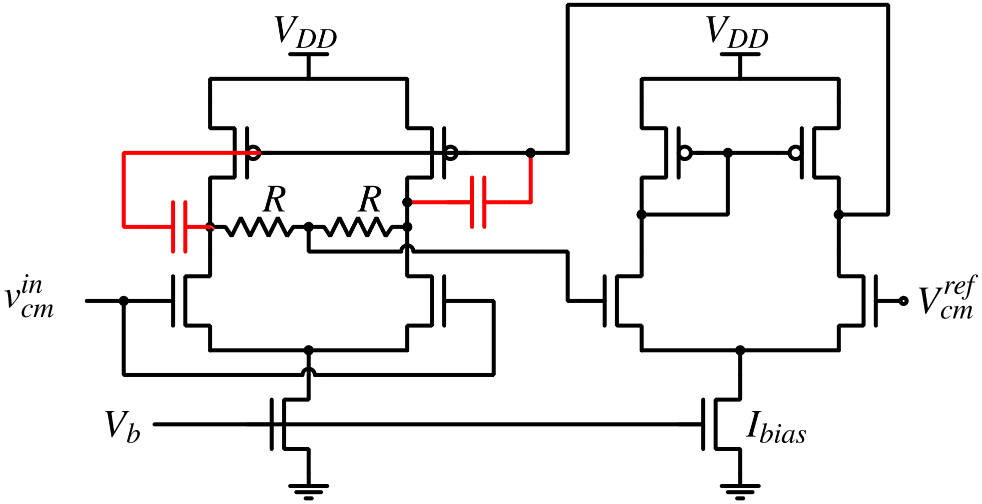

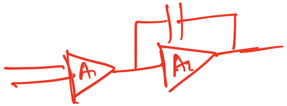

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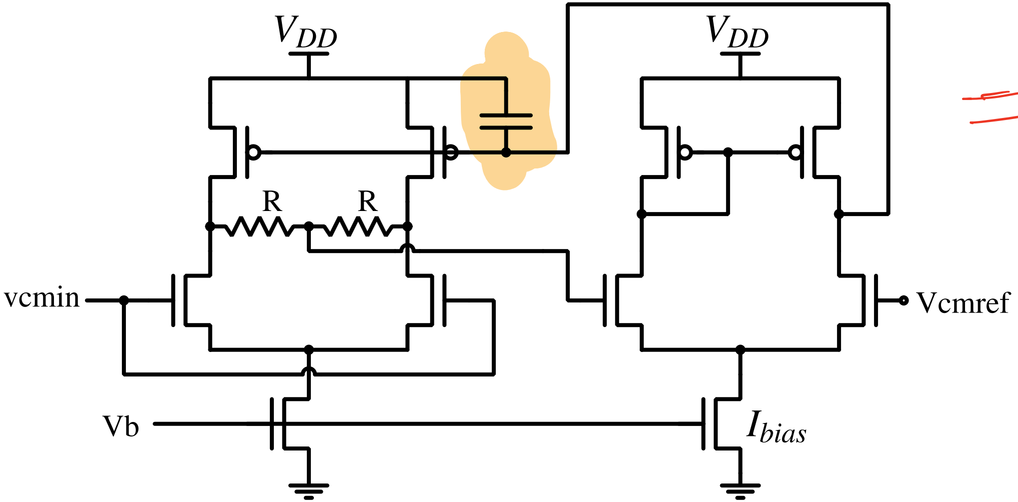
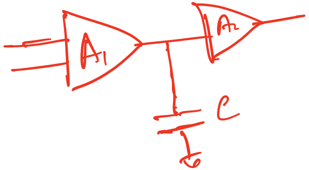
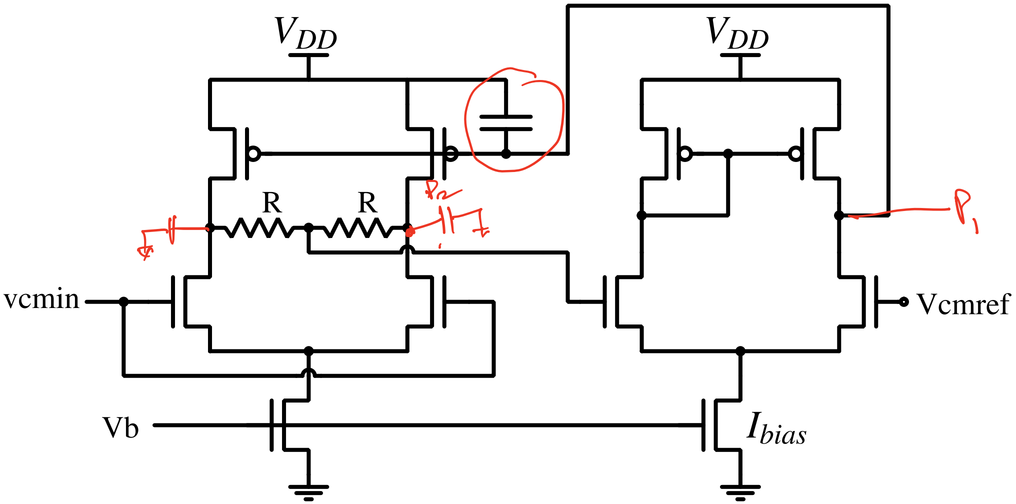
Stabilizing the common mode feedback loop

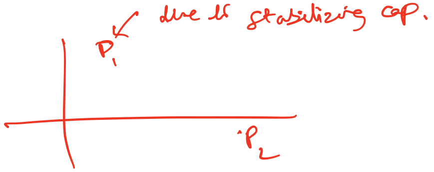
- Hence, we generally do pole splitting using a larger capacitor without using any capacitance amplification of the second stage amplifier of the loop.





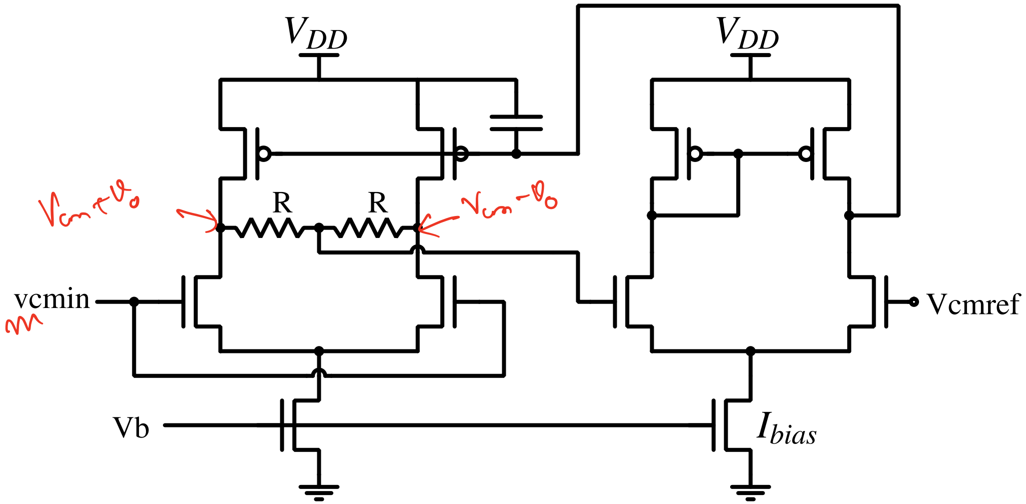
Stabilizing the common mode feedback loop





- Question: Is dynamic performance (bandwidth) of the common mode feedback loop important?

Stabilizing the common mode feedback loop



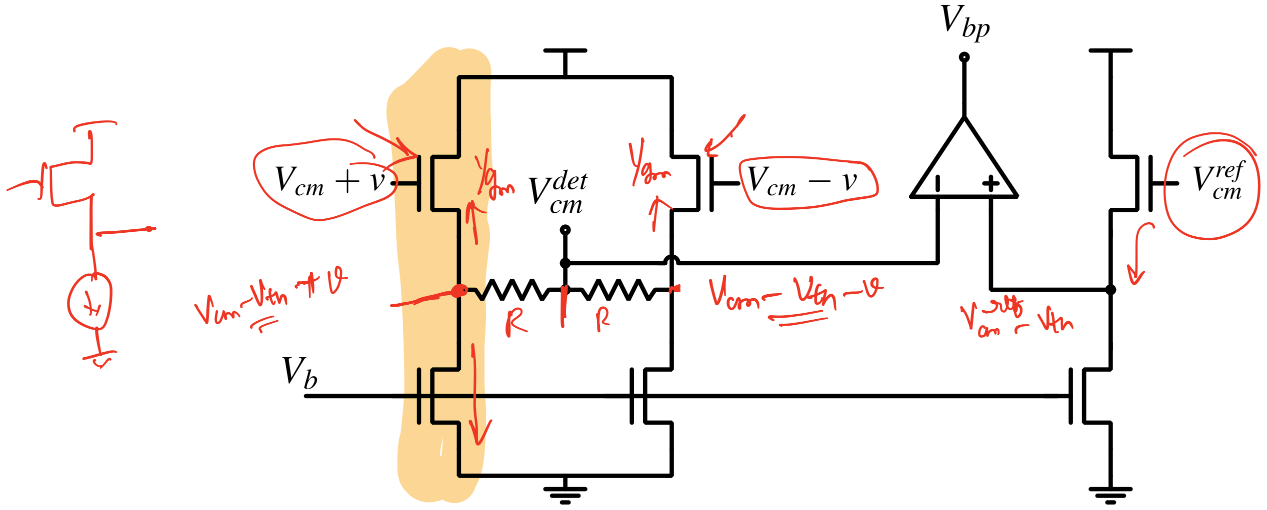
- Question: Is dynamic performance (bandwidth) of the common mode feedback loop important?
- Does the CMRR of the differential amplifier depend on the performance of the CMFB loop?

Common mode detector circuits

- The simple common mode detector using two resistors limits the small signal gain of the amplifier, and using very large resistors is difficult in CMOS process technologies.
- There are other types of common mode detectors which do not need large resistors.
- One could use source follower buffers followed by a resistive CM detector with small resistor values.

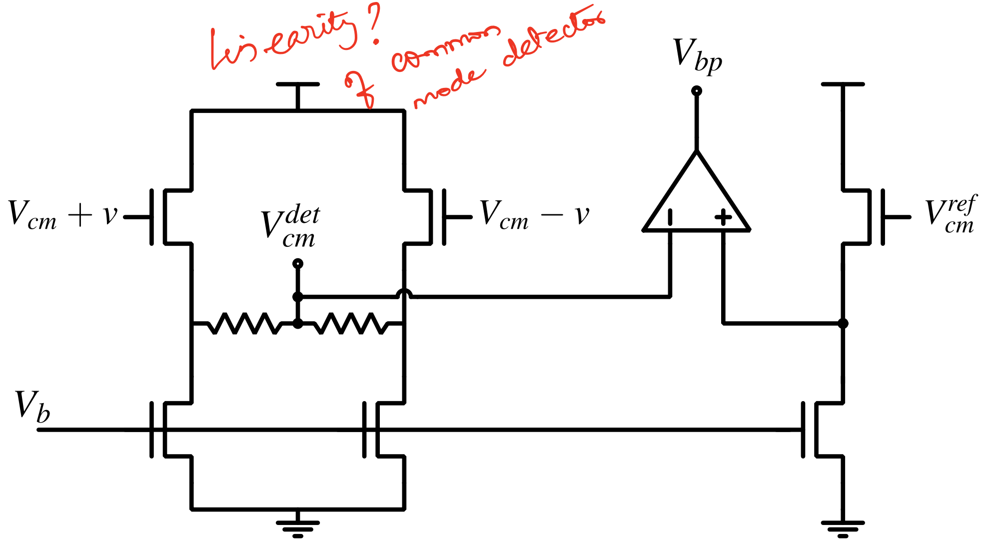
Common mode detector circuits

- One could use source follower buffers followed by a resistive CM detector with small resistor values.



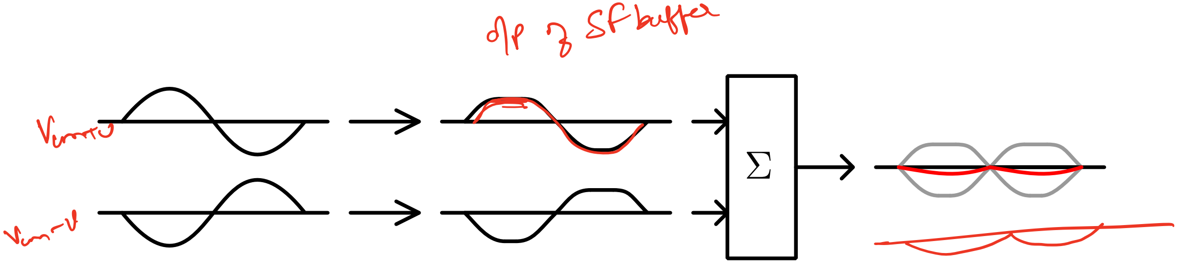
- The source follower introduces a drop (of approximately 1 V_{tn}. Hence, V_{cm}^{ref} is also passed through an identical source follower before the comparison as shown.

Common mode detector circuits

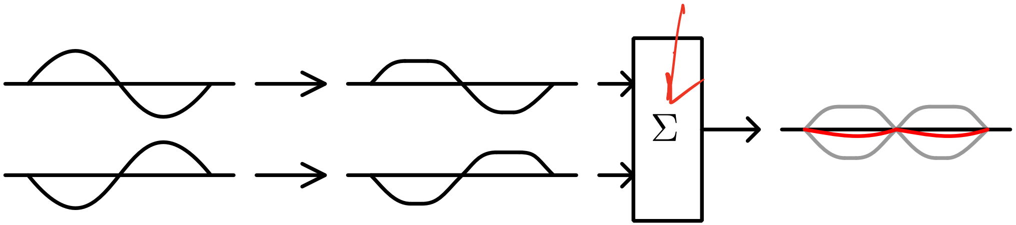


- Linearity of the source follower will affect the dynamic range of the amplifier.

Nonlinearity of the CM detector



Nonlinearity of the CM detector



- The two differential signals can be written as $V_{cm} + v$ and $V_{cm} - v$
- The non-linearity of the CM detector can be generally written as

$$V_{cm}^{det} = \frac{f(V_{cm} + v) + f(V_{cm} - v)}{2}$$

Where $f(x)$ is the non-linearity of the detector (in this case the source follower buffer).

$$V_{cm}^{det} = \frac{f(V_{cm}+v)+f(V_{cm}-v)}{2}$$

$$f(V_{cm}+v) = \alpha_1(V_{cm}+v)+\alpha_2(V_{cm}+v)^2+\alpha_3(V_{cm}+v)^3 \ldots$$

$$f(V_{cm}-v) = \alpha_1(V_{cm}-v)+\alpha_2(V_{cm}-v)^2+\alpha_3(V_{cm}-v)^3 \ldots$$

$$\vdots$$

Nonlinearity of the CM detector

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$$\vdots$$

$$V_{cm}^{det} = \alpha_1 V_{cm} + \alpha_2 V_{cm}^2 + \alpha_2 v^2 + \alpha_3 V_{cm}^3 + 3\alpha_3 V_{cm} v^2 + \dots$$

- Even harmonics (of the small signal input) will appear on the generated bias voltage.

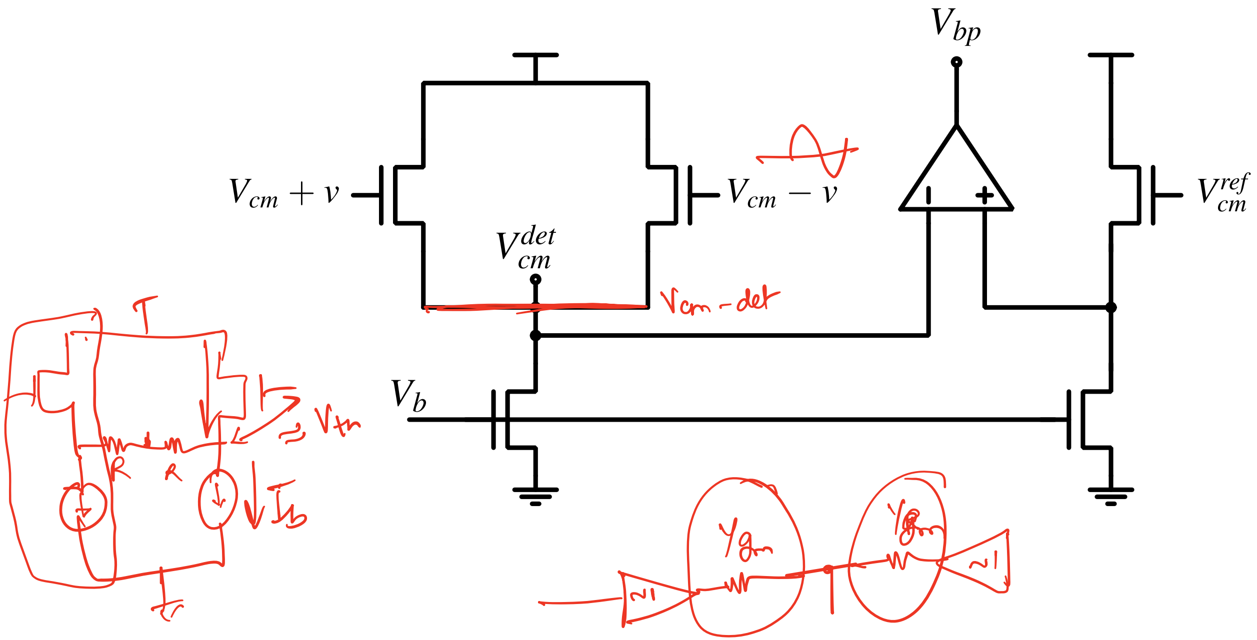
Nonlinearity of the CM detector

$$V_{cm}^{det} = \alpha_1 V_{cm} + \alpha_2 V_{cm}^2 + \alpha_2 v^2 + \alpha_3 V_{cm}^3 + 3\alpha_3 V_{cm} v^2 + \ldots$$

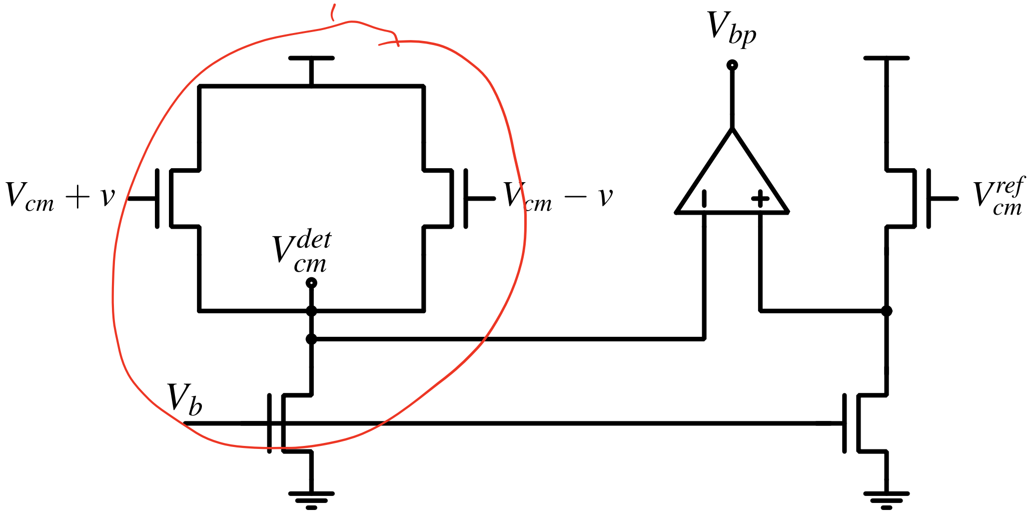
- Ideally V_{cm} is a DC signal. However, note that the harmonics of the small signal appears as a common mode signal at the outputs which results in small signal variations on V_{cm} as well.

- Therefore, linearity of the common mode detector is very important for the performance of the fully differential amplifier.

Common mode feedback using source follower amplifier



Common mode feedback using source follower amplifier

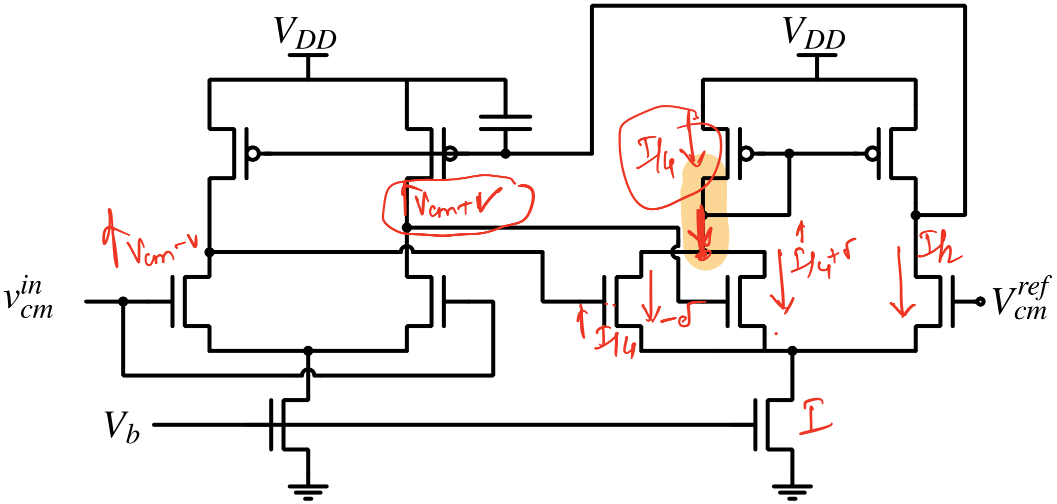


- This source follower amplifier has inferior linearity performance compared to the source follower buffers followed by resistive common mode detector.

Common mode feedback using source follower amplifier

Common mode feedback using a summing amplifier

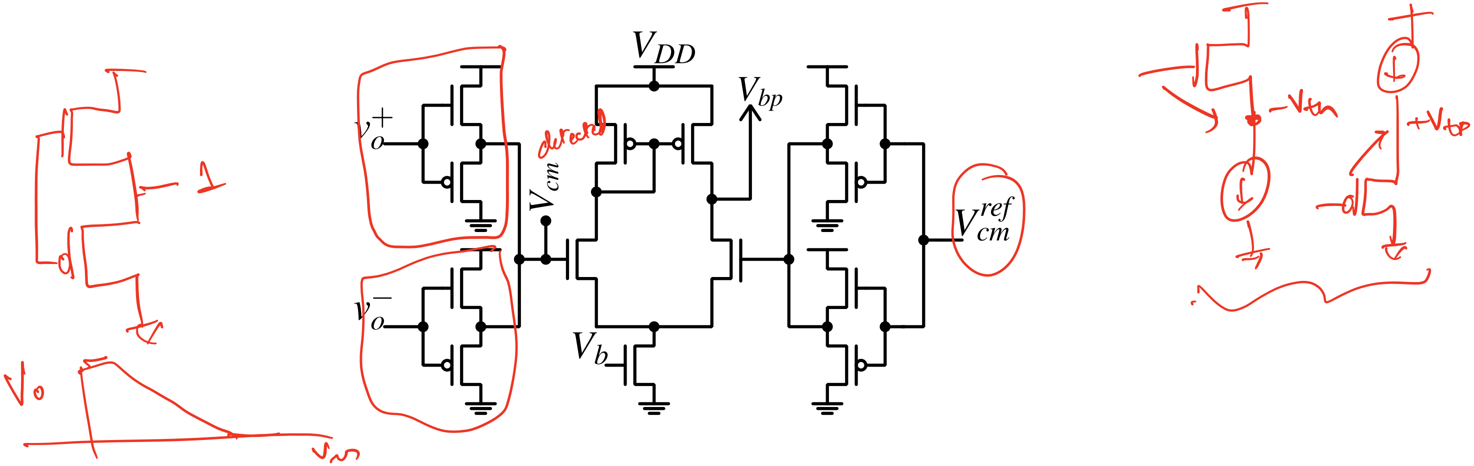
- A summing amplifier can be used as a common mode detector as follows.



$g_m(v) + g_m(-v)$

- This circuit gives high value of gain in the CMFB loop.

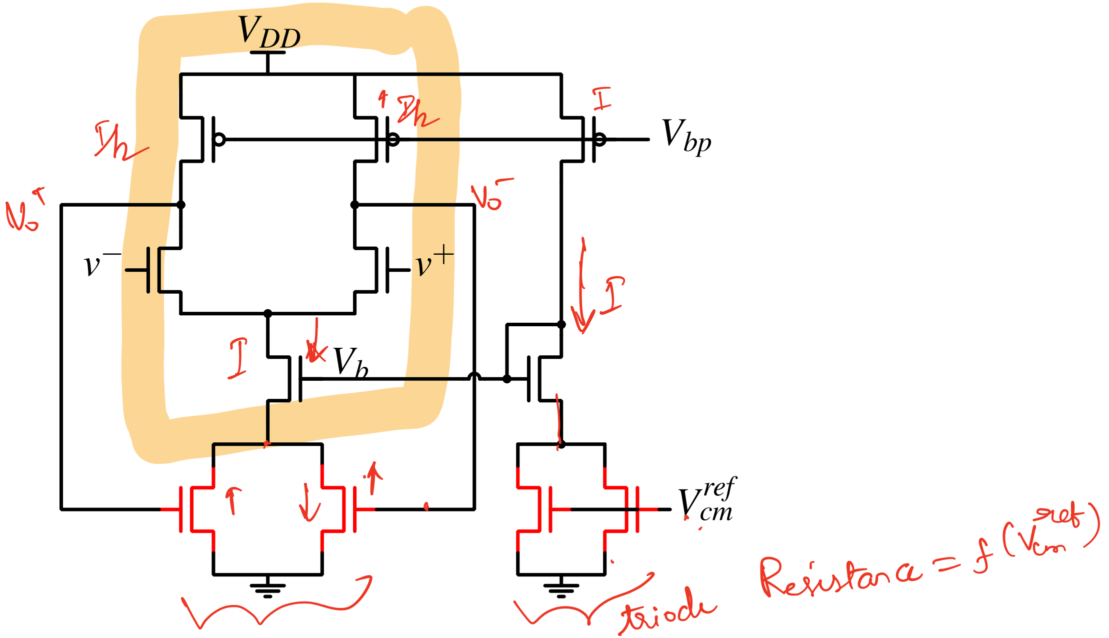
Complementary source follower based CM detector



- This circuit has high linear range due to the use of complementary MOS transistors.
- The DC shift is relatively small $\boxed{(V_{tp} - V_{tn})}$.

M. Daliri et al., “A 0.8-V 420nw cmos switched-opamp switched-capacitor pacemaker front-end with a new continuous-time cmfb," in 2008 15th IEEE Int. Conf. on Electronics, Circ. and Sys., 2008, pp. 758-761.

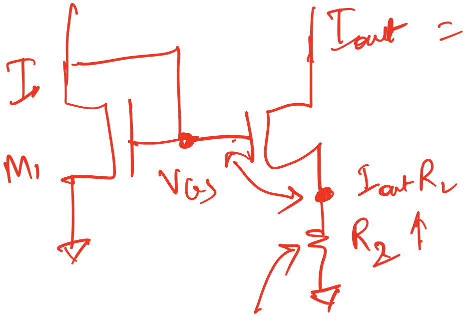
CM detector using transistors in triode region



- The transistors shown in red are biased in triode region.

CM detector using transistors in triode region: Analysis

$$I_{out} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} \left(V_{GS(M_1)} - I_{out} R_2 - V_{th} \right)$$



Questions and Discussions

References

1. Textbook: Paul Gray, Paul Hurst, Stephen Lewis, Robert Meyer, “Analysis and Design of Analog Integrated circuits”
2. Textbook: R. Jacob Baker, “CMOS circuit design, layout and simulation”

Further reading

1. J. F. Duque-Carrillo, “Control of the common-mode component in CMOS continuous-time fully differential signal processing”, Analog Integrated Circuits and Signal Processing, 4(2):131-140, Sep 1993. ISSN 1573-1979. doi: 10.1007/BF01254864